The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Section-Centered Hidden-Response
Body-Lift Response Selection Collapse
to a Canonical Normalized
Selected-Response Lift

Abstract

The current active-primary theorem already proves that the current section-centered displacement-body quadratic forcing law follows canonically from a smaller explicit body-lift response selection $\mathfrak{L}^{\mathrm{disp,body,sel}}$ on the fixed displacement body. The next honest benchmark-facing move is therefore no longer another replay of the body-level law or of the visible-response structures it yields. It is to force the selected-lift package itself, or to prove that some genuinely smaller constituent inside that package already carries all of its benchmark-relevant content.

The present note takes the smaller-constituent route in the cleanest available form. For each sector, a benchmark-faithful body-lift response selection consists of a finite-dimensional selected carrier, a smooth linear selected response lift, and a smooth positive self-adjoint selected response operator acting on that carrier. The theorem shows that this operator-bearing package already collapses canonically to a smaller normalized selected-response lift:

$$\widehat{\Lambda}_{\sigma}^{\mathrm{disp,body,norm}}(y) = K_{\sigma}^{\mathrm{disp,body,sel}}(y)^{1/2}\Lambda_{\sigma}^{\mathrm{disp,body,sel}}(y).$$

The current body-level forcing law is exactly the normalized lift energy

$$\frac{1}{2}\left\langle \widehat{\Lambda}_{\sigma}^{\mathrm{disp,body,norm}}(y)v, \widehat{\Lambda}_{\sigma}^{\mathrm{disp,body,norm}}(y)v \right\rangle$$

on the actual displacement body. Conversely, any benchmark-faithful normalized selected-response lift already induces a benchmark-faithful body-lift response selection by taking the selected response operator to be the identity. Thus the separate operator-bearing presentation carries no additional benchmark-facing freedom beyond the smaller normalized lift.

The benchmark boundary remains unchanged throughout. The one operative metric is still exactly $g^{\sharp}$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. By the current forcing-law theorem, the same displacement-body quadratic forcing law, the same canonical minimal visible-response representative, the same current body-response core, and every currently recorded downstream benchmark-facing consequence then follow unchanged. The remaining honest burden therefore moves below the selected body-lift response package to a deeper derivation or justification of the current normalized selected-response lift itself rather than another same-selection or same-body-law replay.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has now already removed two larger stopping points on the recorded section-centered hidden-response route: first the visible-response operator-family presentation, and then the body-level quadratic forcing law itself. The current active-primary theorem shows that once a smaller explicit body-lift response selection is fixed, the present body-level forcing law is no longer primitive.

The next honest question is therefore narrower again. Does the selected-lift package itself still contain benchmark-relevant freedom, or does it already collapse to some smaller constituent? The present note answers that question in the smallest clean way presently available. It does not introduce a larger current-image-core package, a larger reduced-carrier presentation, an older projection/body presentation, or a replay of the full displacement-response substrate law. Instead it shows that the selected response operator can be absorbed canonically into the lift itself through its positive square root. The benchmark-relevant content of the current selected-lift package is therefore carried already by a smaller normalized selected-response lift on the fixed displacement body.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not claim a completed first-principles derivation of GR from final substrate microphysics. Its narrower positive role is to show that the current operator-bearing body-lift response selection already collapses benchmark-facingly to a smaller canonical normalized selected-response lift.

2. Fixed Inputs from the Current Selected-Lift Frontier

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^{\sharp}, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T^{\mathrm{matter}}_{\mu\nu}[g^{\sharp}, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$.

Definition 2 (Recorded benchmark base and downstream chain)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *The active route already fixes the following data and consequences.*

1. *the current proto-section benchmark base, in particular the recorded proto-section body* $\mathcal{B}^{\mathrm{psec}}_{\sigma}$ *and shell* $\mathcal{Z}^{\mathrm{psec}}_{\sigma}$*;*

2. *the fixed section-centered displacement body*

$$\mathcal{D}^{\mathrm{pppresec}}_{\sigma} \subset \mathcal{B}^{\mathrm{psec}}_{\sigma} \times V^{\mathrm{pppresec}}_{\sigma}$$

   whose fiber

$$\mathcal{D}^{\mathrm{pppresec}}_{\sigma}(y) \equiv \{v \in V^{\mathrm{pppresec}}_{\sigma} : (y, v) \in \mathcal{D}^{\mathrm{pppresec}}_{\sigma}\}$$

   is nonempty, compact, convex, contains 0*, and has nonempty interior for every* $y \in \mathcal{B}^{\mathrm{psec}}_{\sigma}$*;*

3. *the recorded downstream continuation, namely*

$$\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyresp}}, \quad \mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec,res}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec}},$$
$$\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}, \quad \mathfrak{S}^{\mathrm{sub,proto,pppresec,pb}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec,sec}},$$

   together with the current continuity-Hessian compressed core and benchmark one-metric observer package;

4. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Proposition 1 (What the current selected-lift frontier already proves)**.** *At current scope, the already-recorded selected-lift frontier theorems prove all of the following.*

1. *Any benchmark-faithful body-lift response selection canonically determines the current section-centered displacement-body quadratic forcing law on the fixed displacement body.*

2. *That law canonically determines the visible-response operator family* $S^{\mathrm{disp,resp,vis}}_{\sigma}$ *and the canonical minimal visible-response representative* $\mathfrak{S}^{\mathrm{disp,resp,red,min}}_{\sigma}$.

3. *The current body-response core and every currently recorded downstream benchmark-facing consequence then follow unchanged from the same derived representative.*

Proof. Item (1) is exactly the content of the current displacement-body forcing-law theorem from body-lift response selection. Items (2) and (3) are exactly the earlier forcing-law downstream continuation already preserved on the recorded route. □

3. Body-Lift Response Selection

Definition 3 (Benchmark-faithful body-lift response selection)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* benchmark-faithful body-lift response selection *for sector* σ *consists of the following data.*

1. *The fixed proto-section benchmark base and fixed section-centered displacement body of Definition 2(1)–(2).*

2. *A finite-dimensional Euclidean selected-lift carrier*

$$\mathcal{U}_\sigma^{\mathrm{disp,body,sel}}.$$

3. *A smooth family of linear selected response lifts*

$$\Lambda_\sigma^{\mathrm{disp,body,sel}}(y) : V_\sigma^{\mathrm{pppresec}} \to \mathcal{U}_\sigma^{\mathrm{disp,body,sel}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$

   used only on the recorded displacement-body fiber

$$\mathcal{D}_\sigma^{\mathrm{pppresec}}(y).$$

4. *A smooth family of positive self-adjoint selected response operators*

$$K_\sigma^{\mathrm{disp,body,sel}}(y) : \mathcal{U}_\sigma^{\mathrm{disp,body,sel}} \to \mathcal{U}_\sigma^{\mathrm{disp,body,sel}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$

5. *Positivity on the actual displacement body:*

$$\left\langle \Lambda_\sigma^{\mathrm{disp,body,sel}}(y)v, K_\sigma^{\mathrm{disp,body,sel}}(y)\Lambda_\sigma^{\mathrm{disp,body,sel}}(y)v \right\rangle = 0 \qquad \Longleftrightarrow \qquad v = 0$$

   for every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$ and every $v \in \mathcal{D}_\sigma^{\mathrm{pppresec}}(y)$.

6. *Benchmark faithfulness in the sense of Definition 2(4).*

Remark 1. *Definition 3 is already smaller than the older displacement-response substrate-law and projection/body presentations. The question addressed here is whether even this operator-bearing selected-lift package still contains benchmark-relevant freedom once the actual displacement body is fixed.*

4. Canonical Normalized Selected-Response Lift

Definition 4 (Benchmark-faithful canonical normalized selected-response lift)**.** *Fix $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. A* benchmark-faithful canonical normalized selected-response lift *for sector σ consists of the following data.*

1. *The fixed proto-section benchmark base and fixed section-centered displacement body of Definition 2(1)–(2).*

2. *A finite-dimensional Euclidean normalized selected-response carrier*

$$\widehat{\mathcal{U}}_\sigma^{\mathrm{disp,body,norm}}.$$

3. *A smooth family of linear normalized selected-response lifts*

$$\widehat{\Lambda}_\sigma^{\mathrm{disp,body,norm}}(y) : V_\sigma^{\mathrm{pppresec}} \to \widehat{\mathcal{U}}_\sigma^{\mathrm{disp,body,norm}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$

used only on the recorded displacement-body fiber

$$\mathcal{D}_\sigma^{\mathrm{pppresec}}(y).$$

4. *Positivity on the actual displacement body:*

$$\left\langle \widehat{\Lambda}_\sigma^{\mathrm{disp,body,norm}}(y)v, \widehat{\Lambda}_\sigma^{\mathrm{disp,body,norm}}(y)v \right\rangle = 0 \qquad \Longleftrightarrow \qquad v = 0$$

for every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$ and every $v \in \mathcal{D}_\sigma^{\mathrm{pppresec}}(y)$.

5. *Benchmark faithfulness in the sense of Definition 2(4).*

Remark 2. *Definition 4 is strictly smaller than Definition 3. It keeps the response-carrying lift itself, but no longer treats a separate selected response operator as independently primitive data. The operator-bearing presentation is to be recovered only when needed, and in the canonical normalized presentation it becomes the identity.*

Proposition 2 (Square-root normalization produces a canonical normalized lift)**.** *Fix a benchmark-faithful body-lift response selection for sector σ. Define*

$$\widehat{\mathcal{U}}_\sigma^{\mathrm{disp,body,norm}} \equiv \mathcal{U}_\sigma^{\mathrm{disp,body,sel}}, \qquad \widehat{\Lambda}_\sigma^{\mathrm{disp,body,norm}}(y) \equiv K_\sigma^{\mathrm{disp,body,sel}}(y)^{1/2}\Lambda_\sigma^{\mathrm{disp,body,sel}}(y) \tag{2}$$

for each $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$. Then

$$\left(\widehat{\mathcal{U}}_\sigma^{\mathrm{disp,body,norm}}, \widehat{\Lambda}_\sigma^{\mathrm{disp,body,norm}}\right)$$

is a benchmark-faithful canonical normalized selected-response lift.

Proof. Because $K_\sigma^{\mathrm{disp,body,sel}}(y)$ is positive self-adjoint on the finite-dimensional Euclidean space $\mathcal{U}_\sigma^{\mathrm{disp,body,sel}}$, it has a unique positive self-adjoint square root $K_\sigma^{\mathrm{disp,body,sel}}(y)^{1/2}$. Smooth dependence on y follows from the smooth dependence of the positive operator family. Therefore $y \mapsto \widehat{\Lambda}_\sigma^{\mathrm{disp,body,norm}}(y)$ is a smooth family of linear maps.

For $(y, v) \in \mathcal{D}_\sigma^{\mathrm{pppresec}}$,

$$\left\langle \widehat{\Lambda}_\sigma^{\mathrm{disp,body,norm}}(y)v, \widehat{\Lambda}_\sigma^{\mathrm{disp,body,norm}}(y)v \right\rangle = \left\langle \Lambda_\sigma^{\mathrm{disp,body,sel}}(y)v, K_\sigma^{\mathrm{disp,body,sel}}(y)\Lambda_\sigma^{\mathrm{disp,body,sel}}(y)v \right\rangle,$$

so positivity and vanishing only at $v = 0$ on the actual displacement body are exactly Definition 3(5). Benchmark faithfulness is inherited from Definition 3(6). Hence the normalized data satisfy Definition 4. □

Proposition 3 (A normalized selected-response lift induces a body-lift response selection)**.** *Fix a benchmark-faithful canonical normalized selected-response lift for sector σ. Then the data*

$$\mathcal{U}_\sigma^{\mathrm{disp,body,sel}} \equiv \widehat{\mathcal{U}}_\sigma^{\mathrm{disp,body,norm}}, \qquad \Lambda_\sigma^{\mathrm{disp,body,sel}}(y) \equiv \widehat{\Lambda}_\sigma^{\mathrm{disp,body,norm}}(y), \qquad K_\sigma^{\mathrm{disp,body,sel}}(y) \equiv \mathrm{id}_{\widehat{\mathcal{U}}_\sigma^{\mathrm{disp,body,norm}}}$$

form a benchmark-faithful body-lift response selection.

Proof. Smoothness and linearity are immediate. Positivity on the actual displacement body becomes exactly

$$\left\langle \Lambda_\sigma^{\mathrm{disp,body,sel}}(y)v, K_\sigma^{\mathrm{disp,body,sel}}(y)\Lambda_\sigma^{\mathrm{disp,body,sel}}(y)v \right\rangle = \left\langle \widehat{\Lambda}_\sigma^{\mathrm{disp,body,norm}}(y)v, \widehat{\Lambda}_\sigma^{\mathrm{disp,body,norm}}(y)v \right\rangle,$$

which is positive and vanishes only at $v = 0$ by Definition 4(4). Benchmark faithfulness is inherited from Definition 4(5). Therefore the induced data satisfy Definition 3. □

Definition 5 (Normalized selected-lift forcing energy)**.** *Fix $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ and a benchmark-faithful canonical normalized selected-response lift. Define the normalized selected-lift forcing energy on the recorded displacement body by*

$$\mathscr{F}_{\sigma}^{\mathrm{disp,resp,vis,body}}(y, v) \equiv \frac{1}{2} \left\langle \widehat{\Lambda}_{\sigma}^{\mathrm{disp,body,norm}}(y)v, \widehat{\Lambda}_{\sigma}^{\mathrm{disp,body,norm}}(y)v \right\rangle \tag{3}$$

for every $(y, v) \in \mathcal{D}_{\sigma}^{\mathrm{ppresec}}$.

Proposition 4 (The normalized lift energy agrees with the selected-lift energy)**.** *Fix a benchmark-faithful body-lift response selection and let $\widehat{\Lambda}_{\sigma}^{\mathrm{disp,body,norm}}$ be its square-root normalization from Proposition 2. Then for every $(y, v) \in \mathcal{D}_{\sigma}^{\mathrm{ppresec}}$,*

$$\frac{1}{2} \left\langle \widehat{\Lambda}_{\sigma}^{\mathrm{disp,body,norm}}(y)v, \widehat{\Lambda}_{\sigma}^{\mathrm{disp,body,norm}}(y)v \right\rangle = \frac{1}{2} \left\langle \Lambda_{\sigma}^{\mathrm{disp,body,sel}}(y)v, K_{\sigma}^{\mathrm{disp,body,sel}}(y)\Lambda_{\sigma}^{\mathrm{disp,body,sel}}(y)v \right\rangle.$$

Hence the current body-level forcing law determined from the selected-lift package is exactly the normalized lift energy (3)*.*

Proof. This is the identity already used in the proof of Proposition 2. It follows directly from (2) and from self-adjointness of $K_{\sigma}^{\mathrm{disp,body,sel}}(y)^{1/2}$. □

5. Collapsing the Current Selected-Lift Frontier

Theorem 1 (Current body-lift response selection collapses to a canonical normalized selected-response lift)**.** *Fix the primitive continuation data of Definition 1, the recorded benchmark base and downstream chain of Definition 2, and for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ a benchmark-faithful body-lift response selection in the sense of Definition 3. Then the current operator-bearing selected-lift package carries no additional benchmark-facing freedom beyond a smaller canonical normalized selected-response lift. More precisely:*

1. *Proposition 2 canonically associates to the given body-lift response selection a benchmark-faithful normalized selected-response lift $\widehat{\Lambda}_{\sigma}^{\mathrm{disp,body,norm}}$.*

2. *Proposition 3 recovers from that normalized lift a benchmark-faithful body-lift response selection with identity selected response operator, so the separate operator-bearing presentation is not independently primitive at current benchmark scope.*

3. *By Proposition 4, the current section-centered displacement-body quadratic forcing law is exactly the normalized lift energy* (3)*.*

4. *By Proposition 1(2)–(3), the same visible-response operator family $S_{\sigma}^{\mathrm{disp,resp,vis}}$, the same canonical minimal visible-response representative $\mathfrak{S}_{\sigma}^{\mathrm{disp,resp,red,min}}$, the same current body-response core, and every currently recorded downstream benchmark-facing consequence then follow unchanged from that same normalized lift energy.*

5. *The benchmark one-metric package* (1) *is preserved, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

6. *Consequently the remaining honest burden is no longer to restate the operator-bearing body-lift response selection itself, but to derive or justify the smaller canonical normalized selected-response lift from still more primitive substrate structure.*

Proof. Item (1) is Proposition 2. Item (2) is Proposition 3. Item (3) is Proposition 4. Item (4) is exactly the previously recorded downstream continuation applied to the forcing law (3). Item (5) is the benchmark-faithfulness condition built into Definitions 2, 3, and 4. Item (6) is the routing consequence of Items (1)–(5): once the current selected-lift package collapses benchmark-facingly to the smaller normalized lift, another manuscript that merely restates the same operator-bearing selected-lift package or merely replays the same body-law or operator-level consequences does not remove any remaining benchmark-facing freedom. □

Corollary 1 (The operator-bearing selected-lift presentation is no longer primitive)**.** *Under the hypotheses of Theorem 1, the current operator-bearing body-lift response selection is no longer a primitive stopping point on the recorded section-centered hidden-response route. Its benchmark-relevant content is already carried by the smaller canonical normalized selected-response lift.*

Proof. Theorem 1(1)–(3) identifies the same forcing law as the normalized lift energy, while Theorem 1(4) identifies the same downstream benchmark-facing consequences. Therefore the separate operator-bearing selected-lift presentation is dispensable at current benchmark scope. □

6. Conclusion

The positive result of this note is a genuine internal sharpening below the current body-lift response-selection frontier. The present body-lift response selection no longer needs to be treated as a separately primitive operator-bearing package. Its benchmark-relevant content already collapses to a smaller canonical normalized selected-response lift on the fixed displacement body.

The scope remains narrow and honest. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and nothing here upgrades adoption to a completed first-principles derivation of GR from final substrate microphysics.

The next open burden is therefore sharper again. The next benchmark-facing manuscript should derive or justify the current canonical normalized selected-response lift itself from still more primitive explicit substrate structure. Another manuscript that merely restates the same operator-bearing selected-lift package, the same section-centered displacement-body quadratic forcing law, the same visible-response operator family, the same canonical minimal visible-response representative, or a larger same-output current-image-core or reduced-carrier or projection/body or full substrate-law presentation is no longer the clean next target at current scope.